DIGITAL IMAGE PROCESSING COURSE · FINAL PROJECT

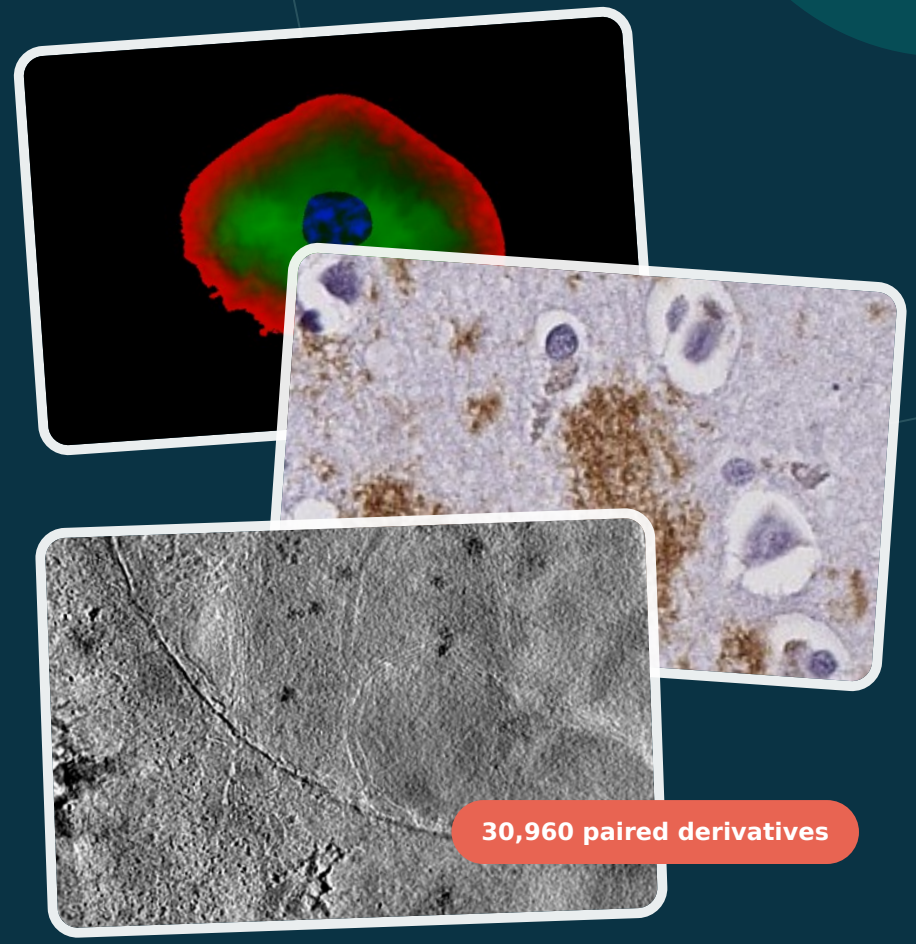
Micro-Q-Instruct

A degradation-aware microscopy benchmark—from controlled classical image processing to paired VLM robustness.

Shayan Salehi
Student ID · 404203509

Ali Hezaveh
Student ID · 404208531

Sharif University of Technology
Department of Computer Engineering · Instructor: Prof. Kasaei



WHY IMAGE QUALITY BELONGS IN MICROSCOPY REASONING

VLMs see the specimen through an acquisition pipeline

> A correct semantic answer may depend on blur, noise, sampling, compression, registration, and detector response—not only biological content.

Motivated by Micro-Bench’s microscopy errors and Q-Instruct’s low-level vision curriculum.

THE MISSING LINK

Quality → evidence → answer

Natural-image quality benchmarks rarely model microscopy physics; microscopy VLM benchmarks rarely isolate low-level corruption.

Our response: a traceable, paired benchmark in which every corruption and restoration is known.

PROJECT THESIS

Three questions, one paired experimental spine

01
Detect

Can an interpretable DIP fingerprint identify degradation family and severity across microscopy domains?

02
Restore

Which matched classical method recovers fidelity, edges, and morphology—and where is information irreversibly lost?

03
Stress-test

Does Qwen3-VL preserve the same correct answer across clean, degraded, and restored views?

Design principle: change the pixels; hold the question, options, answer, model, prompt, and decoding fixed.

OPERATIONAL SYLLABUS ALIGNMENT

Every course chapter becomes an implemented artifact

CH. 2

Digital fundamentals

Sampling, anti-aliasing, quantization, pixel precision

CH. 3

Spatial processing

PSFs, gradients, Laplacian, gamma, CLAHE

CH. 4

Frequency domain

Spectral energy, sinusoidal interference, notch rejection

CH. 5

Restoration

Poisson/read noise, local Wiener, deconvolution

CH. 7

Transforms

Wavelet energy fingerprints and denoising

CH. 8

Compression

Controlled JPEG and block-boundary evidence

CH. 9

Morphology

Otsu proxy masks, Dice/IoU, components

BRIDGE

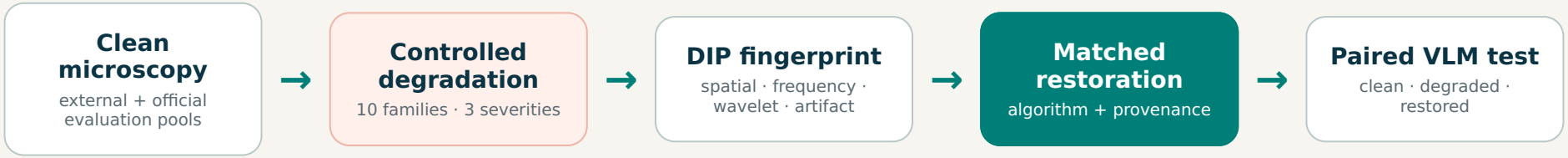
Vision-language robustness

Paired inference, uncertainty, exact-answer scoring

Foundations: Micro-Bench (NeurIPS 2024) · Q-Instruct (CVPR 2024) · Q-Bench (ICLR 2024)

FROM PIXELS TO ROBUSTNESS EVIDENCE

The project is a two-phase causal chain



PHASE 1

Classical DIP benchmark

Produces paired images, exact recipes, interpretable measurements, morphology metrics, restorations, and a classical baseline.

PHASE 1.5

Inference-time robustness

Measures answer stability with paired statistics; no model training or fine-tuning occurs.

STRICT TWO-POOL POLICY

Train on permissive sources; evaluate only on pinned Micro-Bench

1,701
unique clean sources

→

804
training · BBBC010, BBBC038, IDR0067

147
development · held out for calibration

750
evaluation · 15 official Micro-Bench datasets

Contamination guard: decoded-pixel SHA-256 + 64-bit perceptual-hash nearest-neighbor audit + license allowlist.

Source composition



Licenses retained per record: CC0 / CC0-1.0 / CC-BY-4.0 / CC-BY-SA-4.0. Micro-Bench revision pinned to **3f2c5b59…**

BALANCED, DETERMINISTIC DEGRADATION PLANNING

15,480 recipes become 30,960 paired derivatives

10	3	15,480	15,480
DEGRADATION FAMILIES	SEVERITY LEVELS	DEGRADED IMAGES	RESTORED IMAGES

Balanced incomplete factorial

Nine variants per source select three applicable families × weak/medium/strong. Deterministic clean controls and mixed corruptions are added at configured fractions.

MASTER SEED · 20260724

Physics-aware applicability

Photobleaching and channel shift require fluorescence; chromatic shift requires multichannel content; defocus is restricted to optical modalities.

PARAMETERS SCALE WITH IMAGE SIZE

Every recipe preserves image role, exact parameters, seed, bit depth, SHA-256, source license, and output path.

INTERPRETABLE EVIDENCE—NOT A BLACK BOX

Each image yields a fingerprint, paired metrics, and a matched baseline

19-feature fingerprint

Histogram, entropy, gradient/Laplacian, Fourier ratios, spectral centroid, wavelet energy, blockiness, clipping, and channel shift.

Paired image evidence

MSE, PSNR, SSIM, edge retention, histogram JS divergence; Dice/IoU, area, and components where masks exist.

Strategy registry

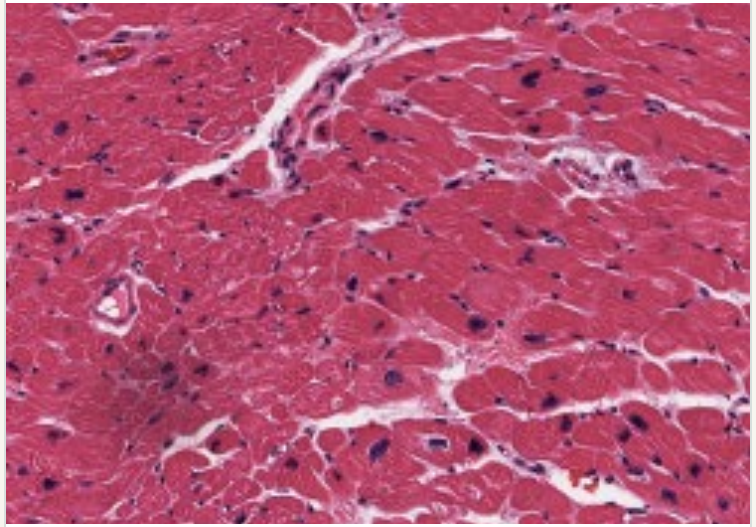
Local Wiener, PSF-aware deconvolution, notch rejection, channel registration, gamma/CLAHE, and wavelet comparisons.

Validated release: 15,480 analyses, 0 errors, 0 warnings; 3,743 / 3,745 eligible severity triplets are PSNR-monotonic.

Sampling / resolution loss

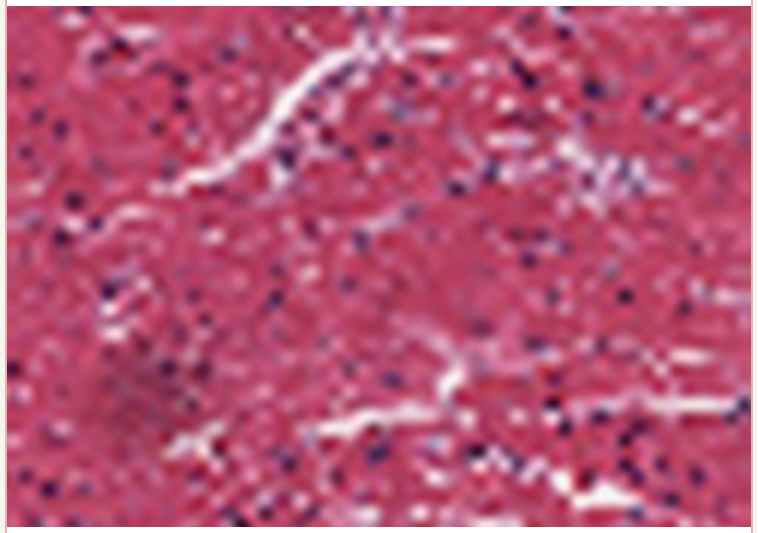
STRONG SEVERITY

01 Clean reference



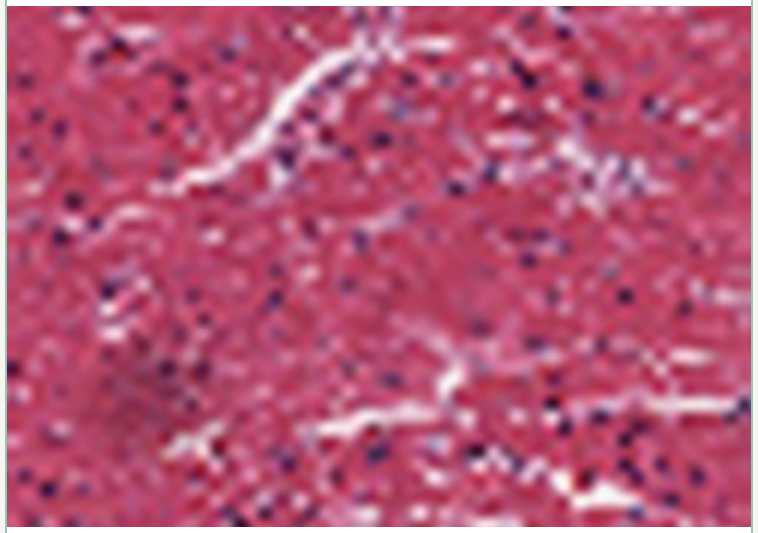
Original · Light

02 Degraded



PSNR 19.6 dB · SSIM 0.414

03 Restored



ΔPSNR +0.0 dB · ΔSSIM -0.000

DEGRADE Downsample then reconstruct at the original dimensions; the strong recipe removes spatial samples and exposes aliasing/blur.

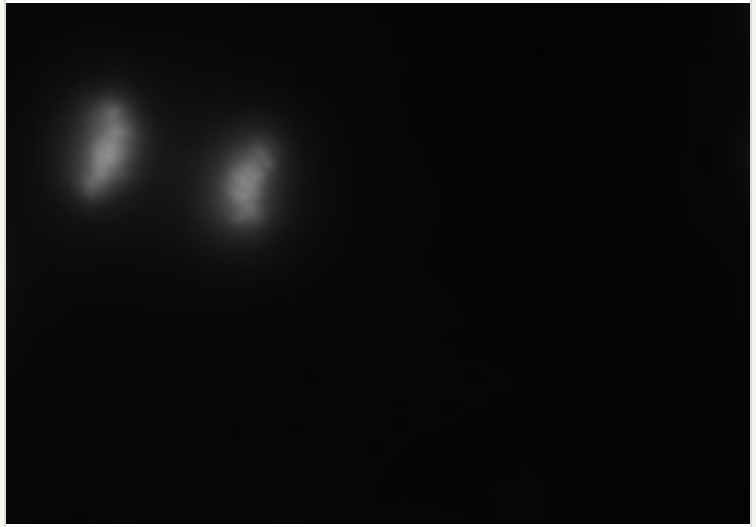
RESTORE BayesShrink db2 wavelet denoising suppresses artifacts, but removed samples are not invertible.

Illustrative evaluation record · nirschl · et · al · 2018 · recipe 763e5f2e

Bit-depth quantization

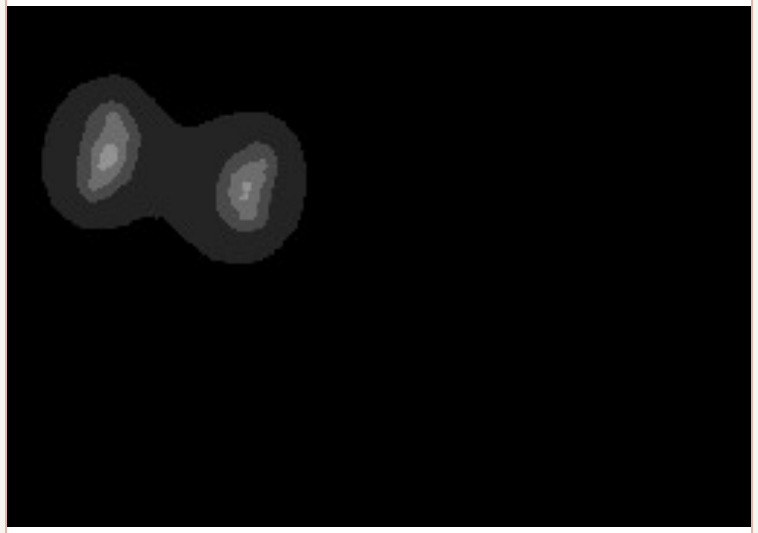
STRONG SEVERITY

01 Clean reference



Original · Fluorescence

02 Degraded



PSNR 30.7 dB · SSIM 0.193

03 Restored



ΔPSNR +0.4 dB · ΔSSIM +0.008

DEGRADE Uniform intensity quantization collapses the continuous tonal range into a small set of discrete levels.

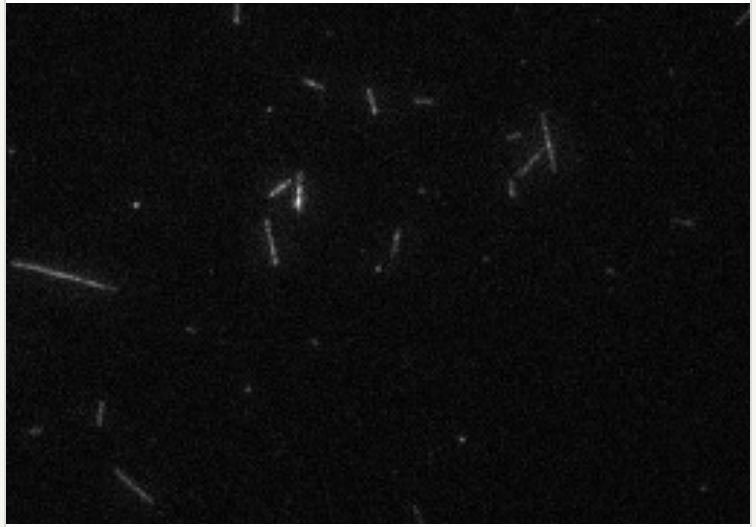
RESTORE Wavelet denoising softens contouring, but it cannot recreate discarded intensity levels.

Illustrative evaluation record · nirschl · unpub · fluorescence · recipe 138721ed

Poisson–Gaussian noise

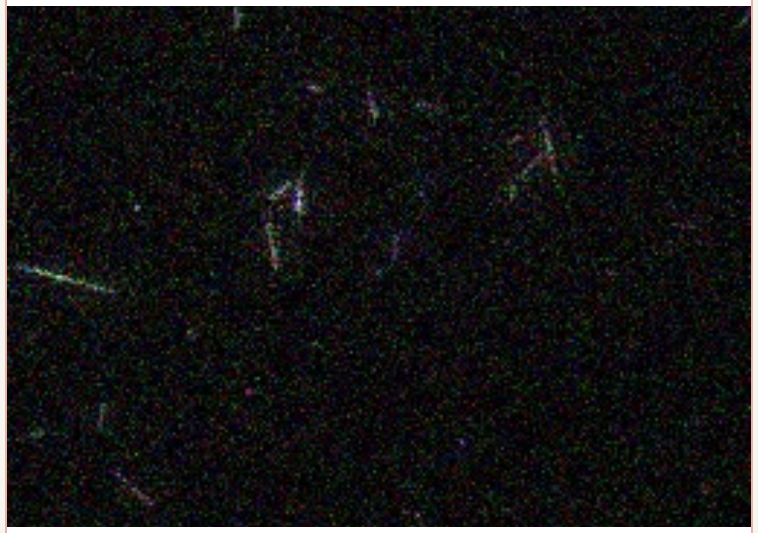
STRONG SEVERITY

01 Clean reference



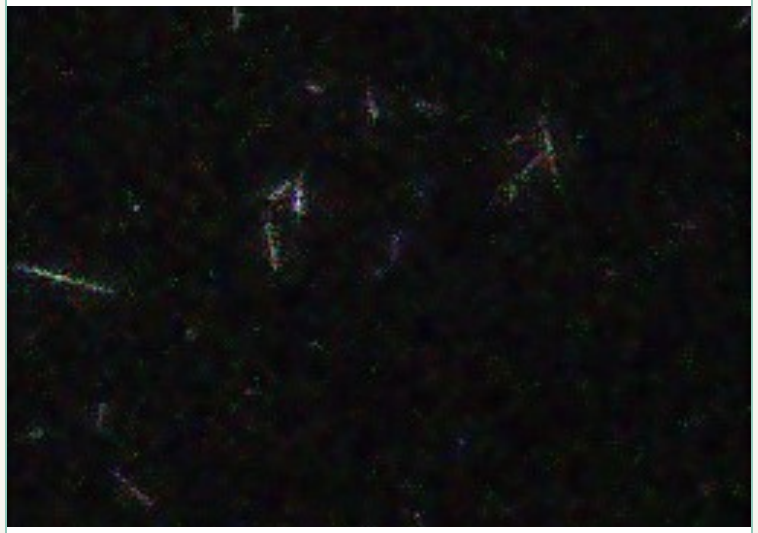
Original · Fluorescence

02 Degraded



PSNR 23.3 dB · SSIM 0.282

03 Restored



ΔPSNR +6.9 dB · ΔSSIM +0.343

DEGRADE Photon-count shot noise is combined with additive read noise to mimic noisy acquisition.

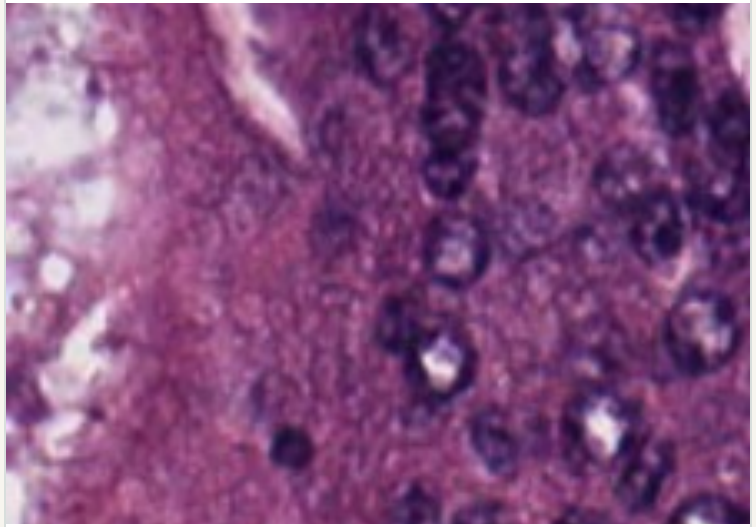
RESTORE A local adaptive Wiener filter estimates neighborhood statistics and attenuates signal-dependent noise.

Illustrative evaluation record · nirschl · unpub · fluorescence · recipe 97e1af78

Defocus blur

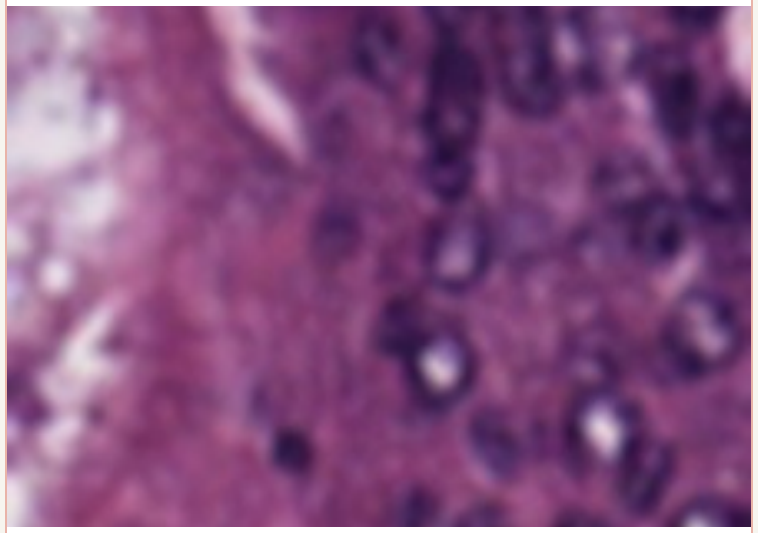
STRONG SEVERITY

01 Clean reference



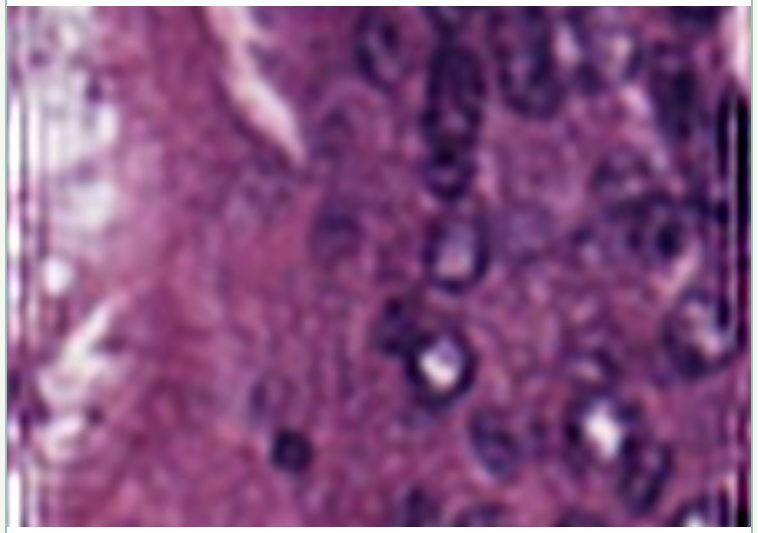
Original · Light

02 Degraded



PSNR 26.7 dB · SSIM 0.700

03 Restored



ΔPSNR +0.6 dB · ΔSSIM +0.144

DEGRADE A normalized disk point-spread function removes high-frequency focus cues without assigning focus direction.

RESTORE Recorded-PSF Wiener deconvolution restores detail while trading sharpness against ringing and noise amplification.

Illustrative evaluation record · kather · et · al · 2016 · recipe d8df307a

Motion / scan drift

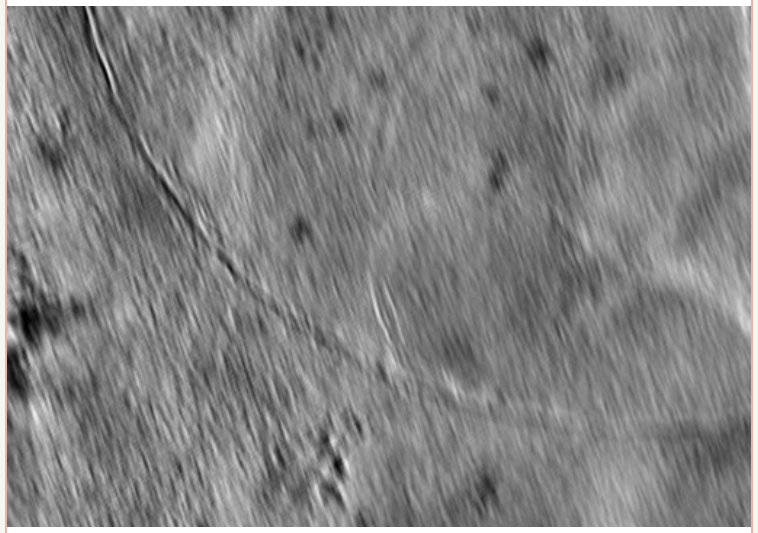
STRONG SEVERITY

01 Clean reference



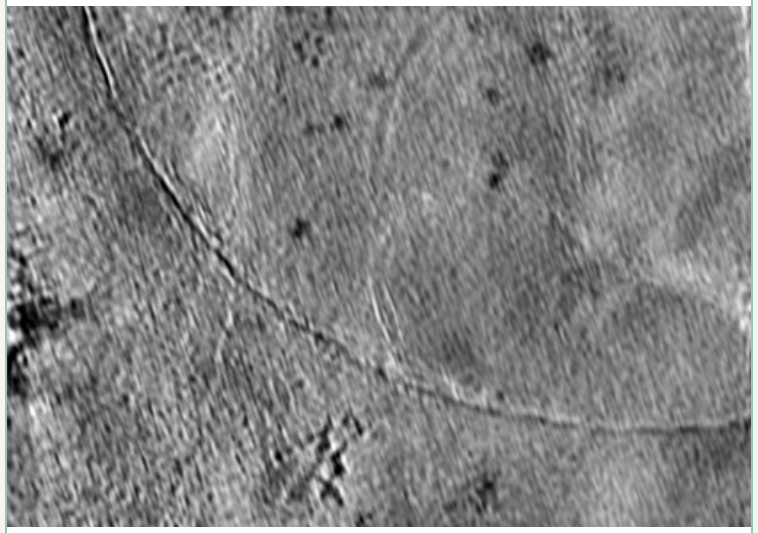
Original · Electron

02 Degraded



PSNR 18.7 dB · SSIM 0.374

03 Restored



ΔPSNR +1.8 dB · ΔSSIM +0.252

DEGRADE A normalized linear PSF smears structures along a controlled length and angle.

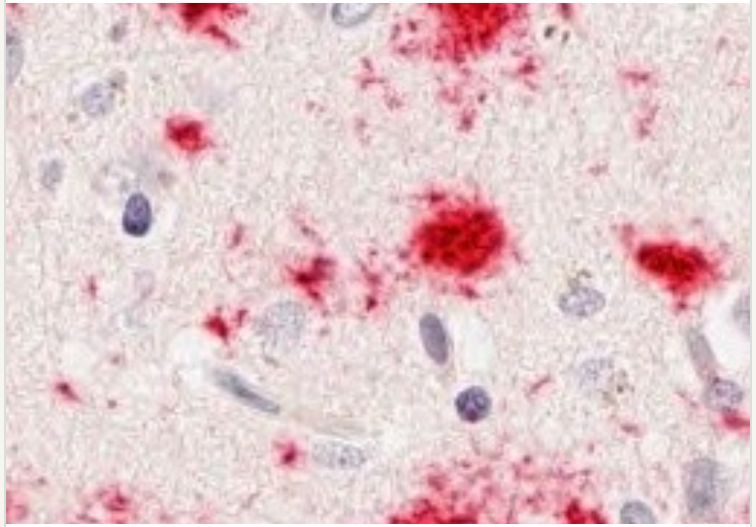
RESTORE Recorded linear-PSF Wiener deconvolution partially reconstructs edge direction and local texture.

Illustrative evaluation record · wu · et · al · 2023 · recipe 88d2ff2e

Periodic interference

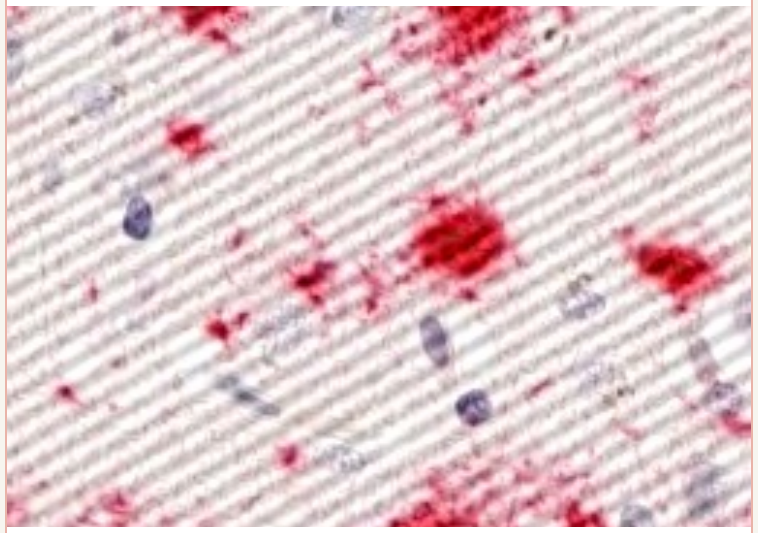
STRONG SEVERITY

01 Clean reference



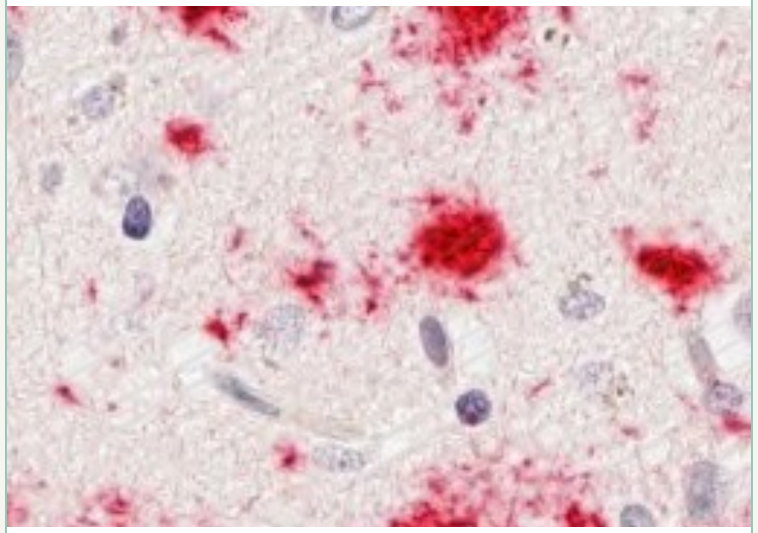
Original · Light

02 Degraded



PSNR 21.8 dB · SSIM 0.503

03 Restored



ΔPSNR +10.5 dB · ΔSSIM +0.431

DEGRADE An oriented sinusoid adds coherent striping at a known spatial frequency and amplitude.

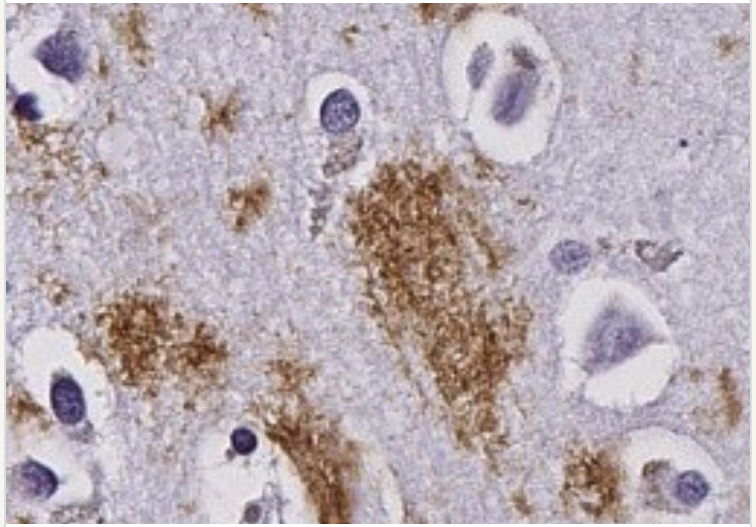
RESTORE A paired notch-reject filter removes the synthetic frequency peaks—an explicitly oracle-assisted baseline.

Illustrative evaluation record · wong · et · al · 2022 · recipe 0202f727

JPEG compression

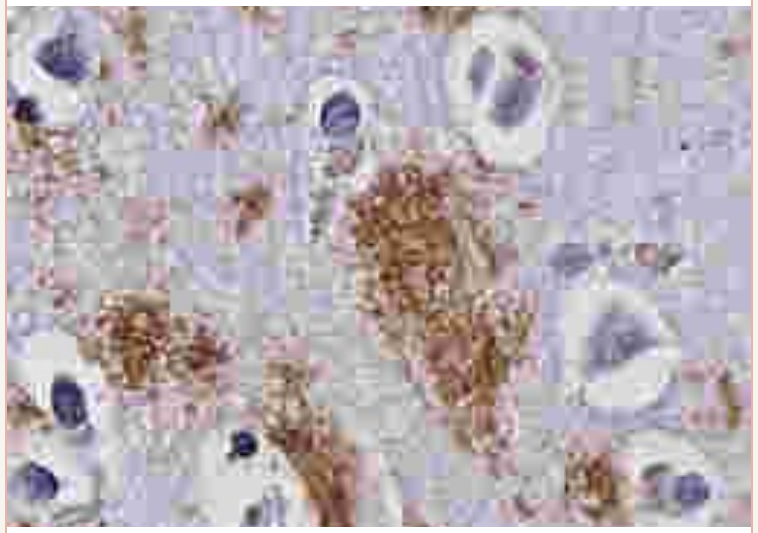
STRONG SEVERITY

01 Clean reference



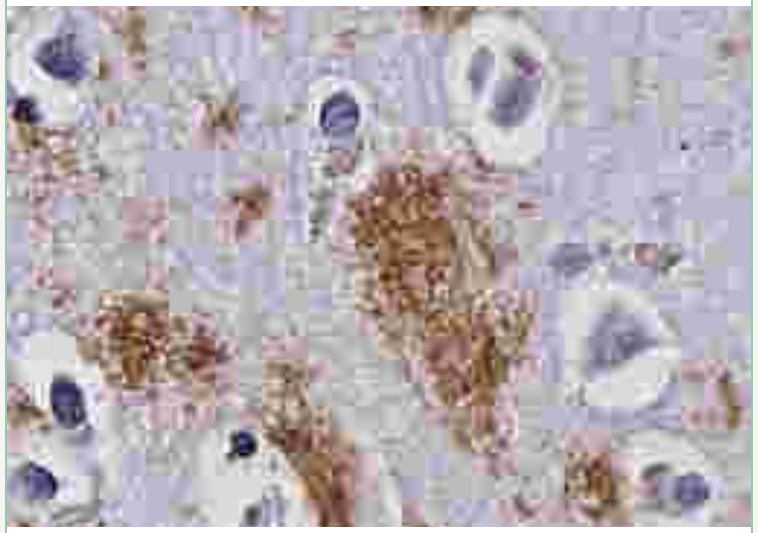
Original · Light

02 Degraded



PSNR 23.2 dB · SSIM 0.639

03 Restored



ΔPSNR +0.0 dB · ΔSSIM +0.000

DEGRADE Lossy block transform coding discards coefficients and introduces block/ringing artifacts; 16-bit inputs also incur an 8-bit round trip.

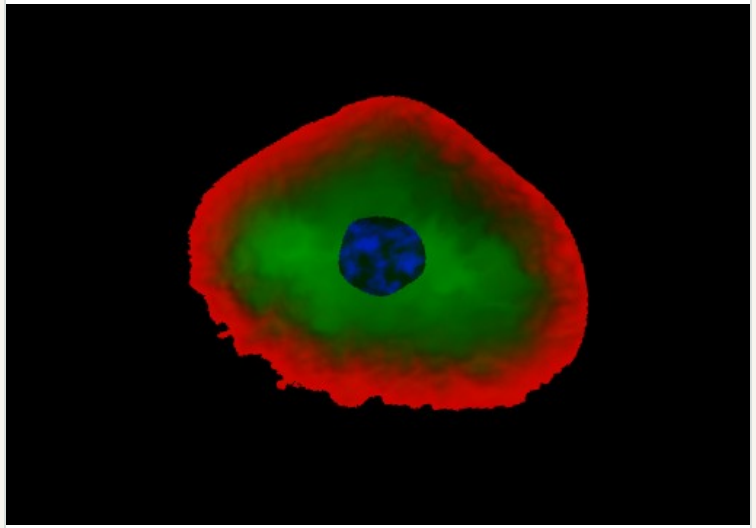
RESTORE Wavelet denoising compares post-compression smoothing, but discarded coefficients remain unrecoverable.

Illustrative evaluation record · tang · et · al · 2019 · recipe c0c1988b

Photobleaching

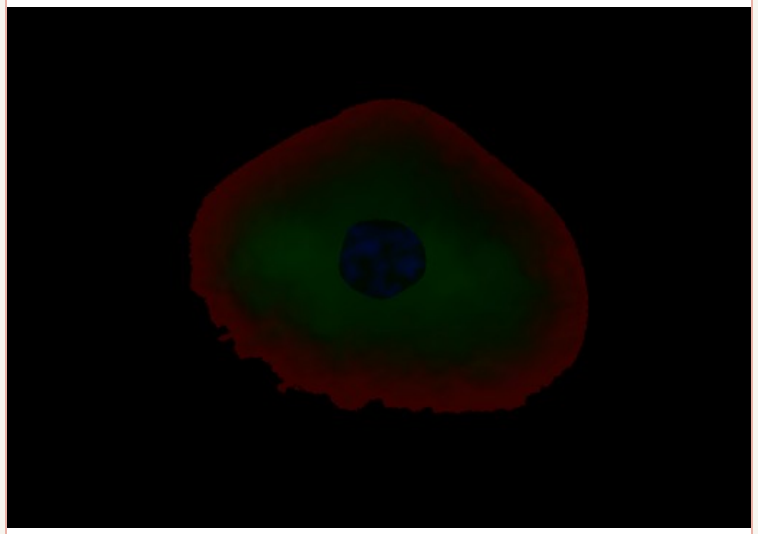
STRONG SEVERITY

01 Clean reference



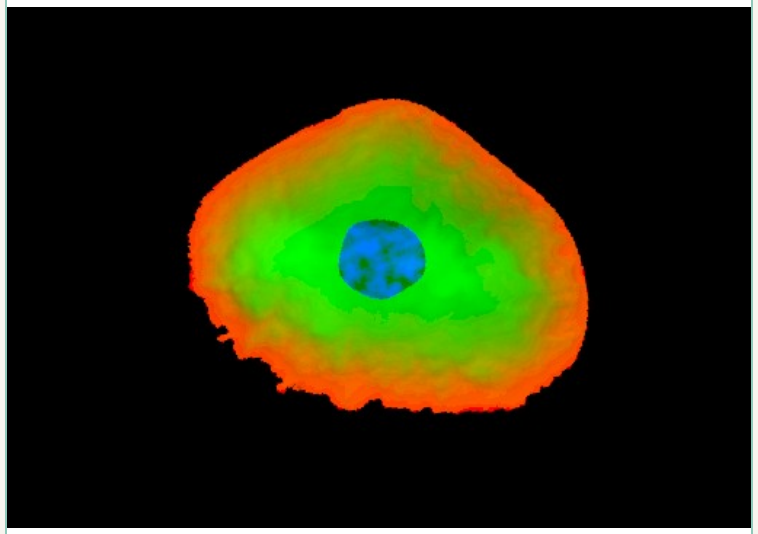
Original · Fluorescence

02 Degraded



PSNR 21.4 dB · SSIM 0.930

03 Restored



ΔPSNR -2.5 dB · ΔSSIM +0.015

DEGRADE Fluorescence signal is attenuated above background to model retained intensity—not detector clipping.

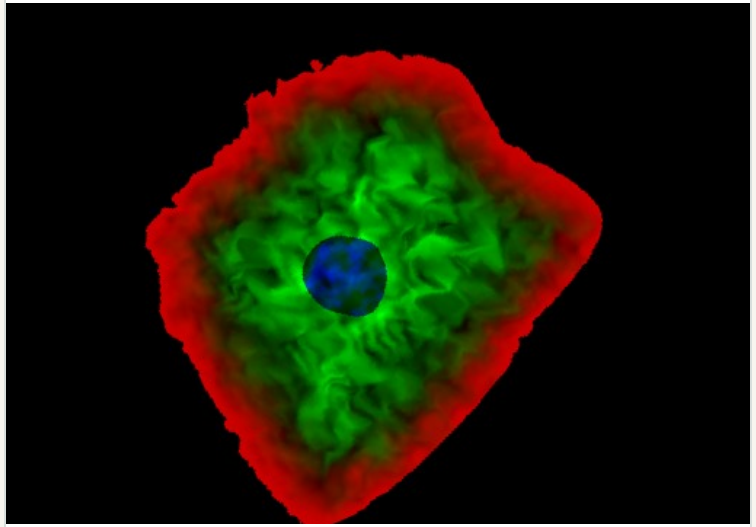
RESTORE Gamma correction plus CLAHE restores visibility and local contrast without recreating lost photons.

Illustrative evaluation record · burgess · et · al · 2024 · contour · recipe 8f1759a7

Chromatic misregistration

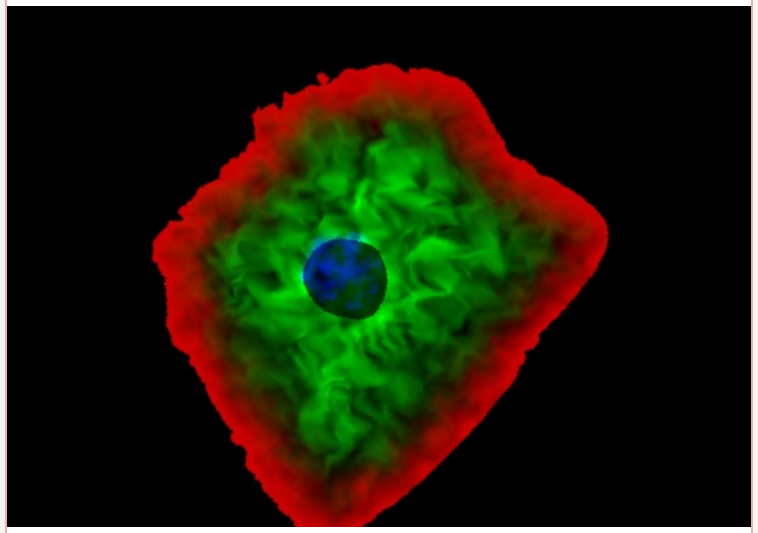
STRONG SEVERITY

01 Clean reference



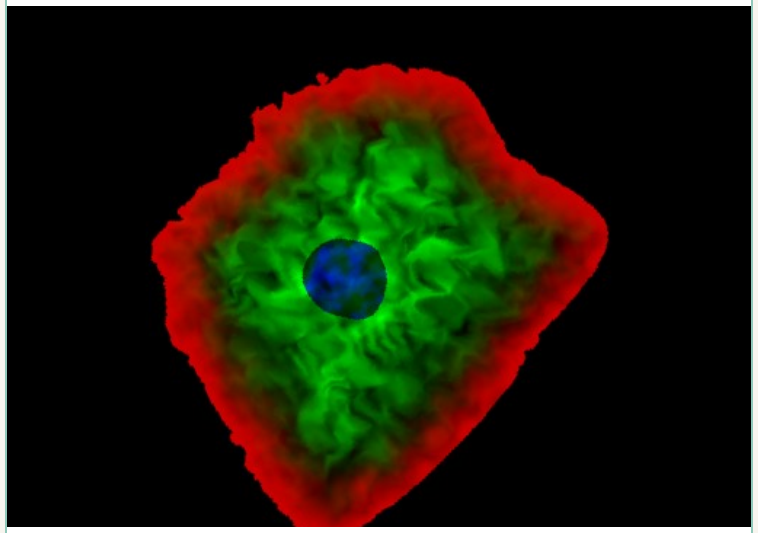
Original · Fluorescence

02 Degraded



PSNR 23.9 dB · SSIM 0.963

03 Restored



ΔPSNR +0.3 dB · ΔSSIM +0.005

DEGRADE Opposing subpixel channel shifts create color fringes in genuinely multichannel fluorescence images.

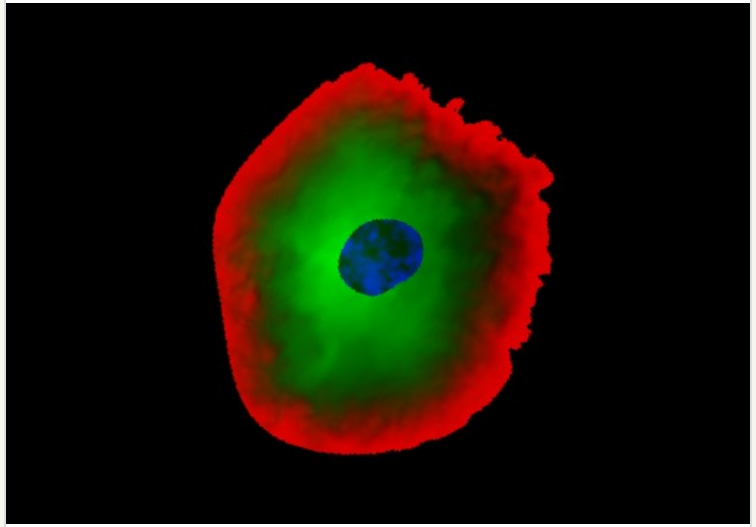
RESTORE Phase-correlation channel registration estimates displacement relative to green and realigns red/blue channels.

Illustrative evaluation record · burgess · et · al · 2024 · contour · recipe 23138828

Detector saturation

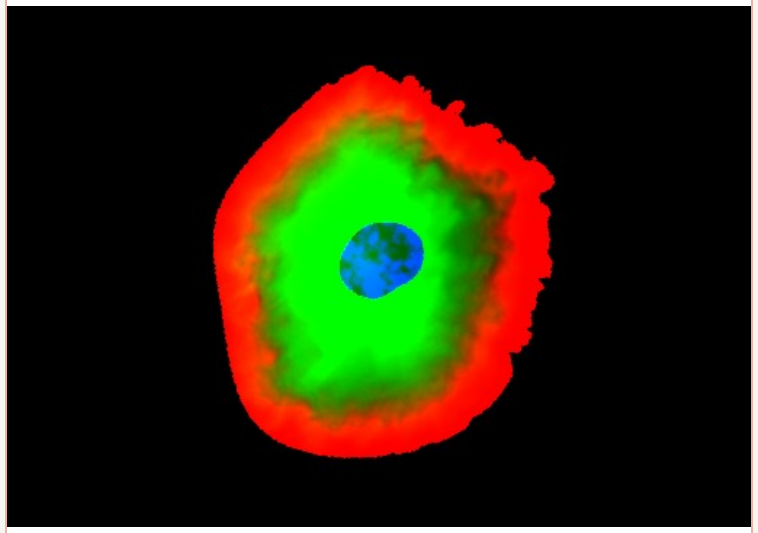
STRONG SEVERITY

01 Clean reference



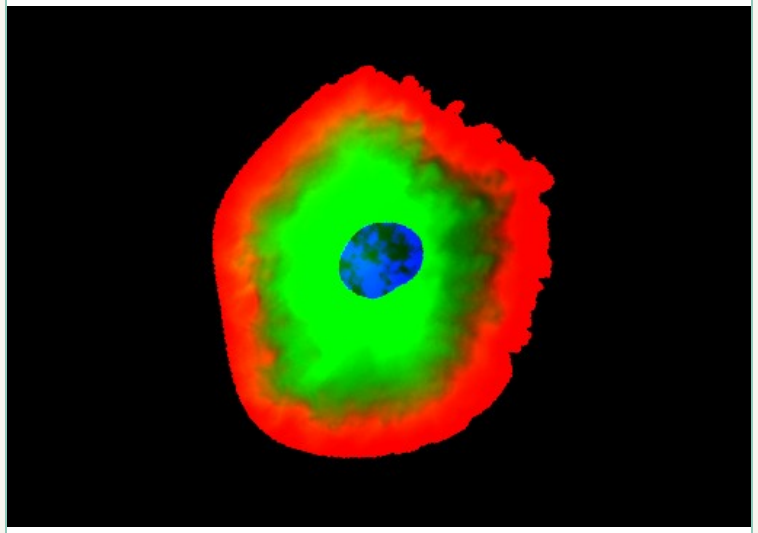
Original · Fluorescence

02 Degraded



PSNR 20.3 dB · SSIM 0.957

03 Restored



ΔPSNR +0.1 dB · ΔSSIM +0.001

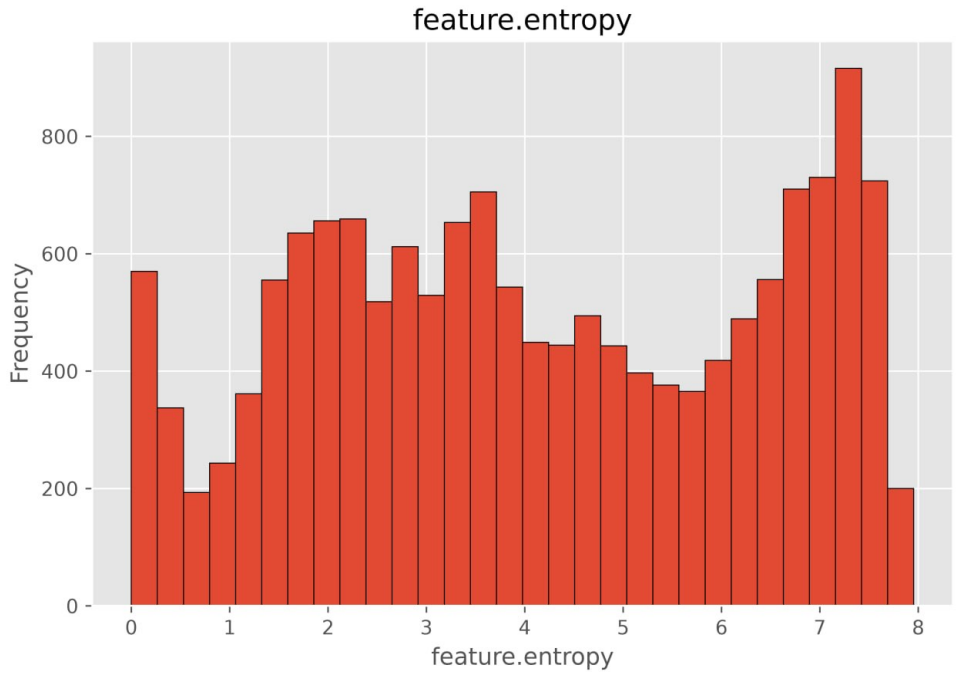
DEGRADE Gain and black-level changes are followed by detector-range clipping, flattening bright structures.

RESTORE Gamma/CLAHE enhancement redistributes visible contrast; clipped values cannot be reconstructed.

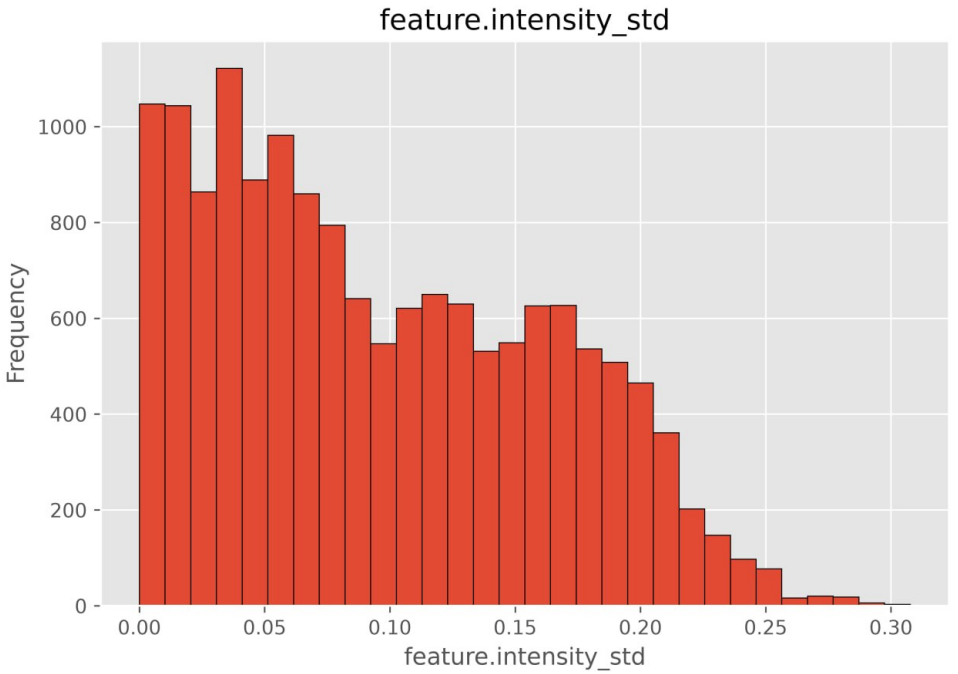
Illustrative evaluation record · burgess · et · al · 2024 · texture · recipe 6120d3a5

FINGERPRINT DISTRIBUTIONS

Feature space spans radically different microscopy regimes



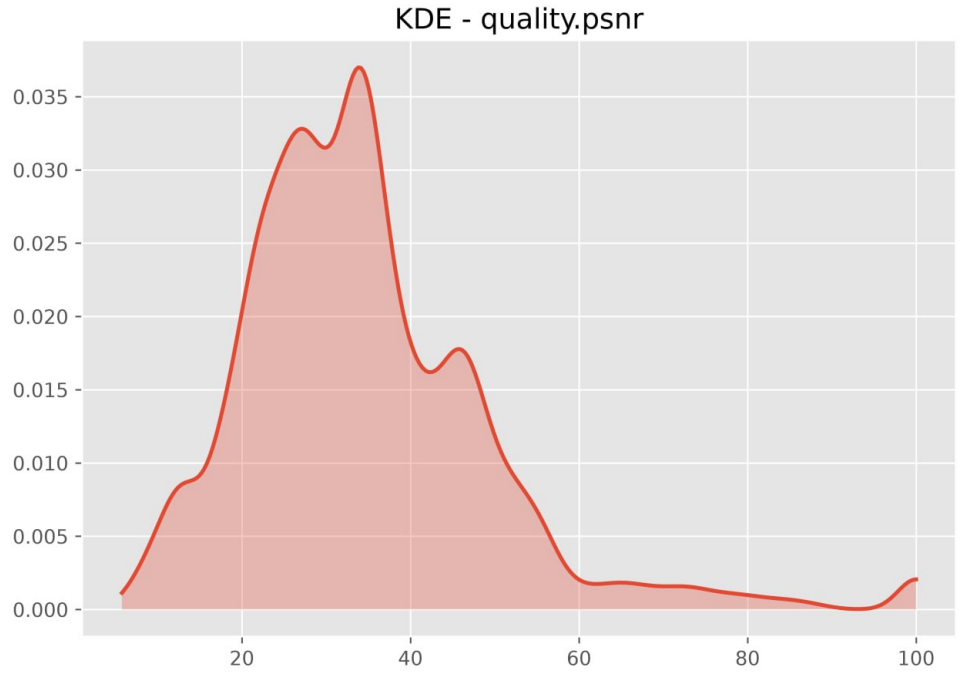
Entropy is broad and multimodal: low-information fields coexist with richly textured specimens, so one global threshold would be brittle.



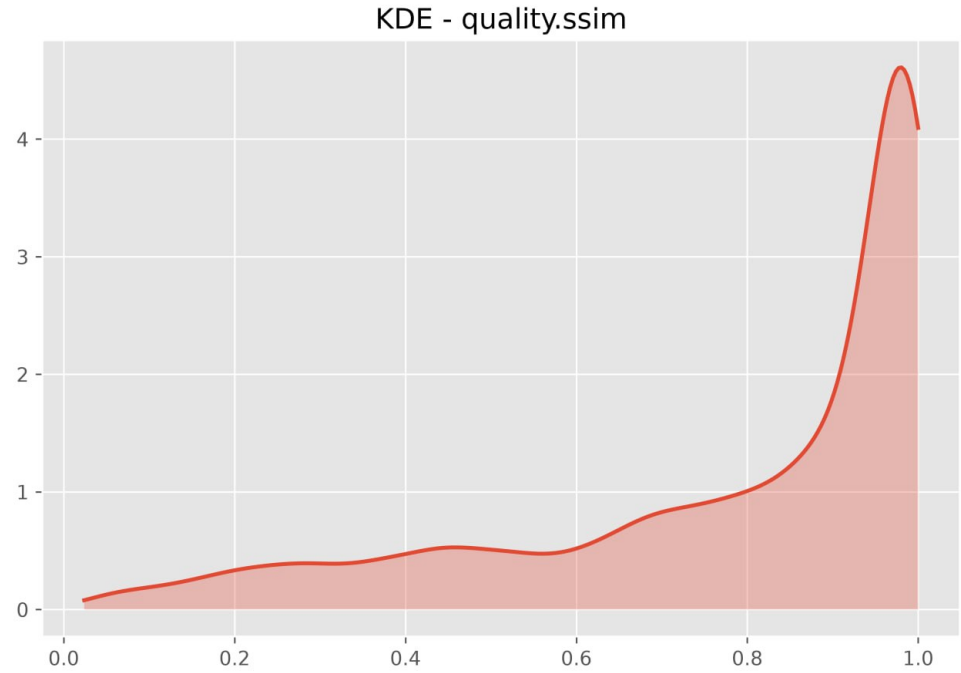
Intensity variation is right-skewed: most images have modest contrast, while a small high-contrast tail motivates normalization and modality-aware interpretation.

PAIRED FIDELITY DISTRIBUTIONS

One scalar cannot summarize degradation impact



PSNR concentrates near 25–40 dB but retains long tails, reflecting both subtle controls and destructive strong corruptions.



SSIM piles up near 1.0 while still spanning the full range; structural similarity and pixel error capture complementary failure modes.

WHAT THE COMPLETED OUTPUTS ESTABLISH

Scale is useful only when the audit trail survives

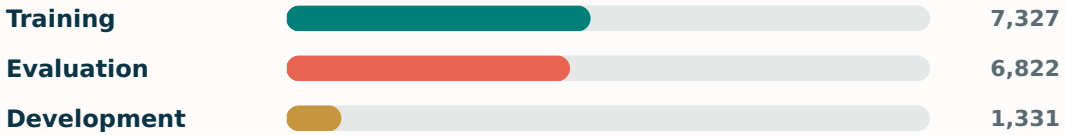
1,701
CLEAN SOURCE IMAGES

15,480
MATERIALIZED RECIPES

99.95%
PSNR MONOTONIC TRIPLETS

0 / 0
VALIDATION ERRORS / WARNINGS

Role-balanced records



Release anatomy

- recipe / generated / restored JSONL manifests
- paired CSV + JSONL analysis records
- dataset card, audit, validation, calibration figures
- structured logs + serialized classical baselines

RESTORATION IS FAMILY-DEPENDENT

Invertibility—not visual polish—governs recovery

Mean evaluation Δ after restoration · single degradations

FAMILY	ΔPSNR	ΔSSIM	INTERPRETATION
Periodic	**+7.79 dB**	**+0.109**	known-frequency notch excels
Poisson–Gaussian	**+5.21 dB**	**+0.165**	local statistics are recoverable
Motion	**+1.43 dB**	**+0.044**	partial PSF inversion
Quantization	**−2.59 dB**	**−0.038**	levels were discarded
Photobleaching	**−12.11 dB**	**−0.274**	enhancement ≠ photon recovery

CROSS-SOURCE FINGERPRINT BASELINE

34.9%

degradation-family accuracy · macro-F1 27.3%

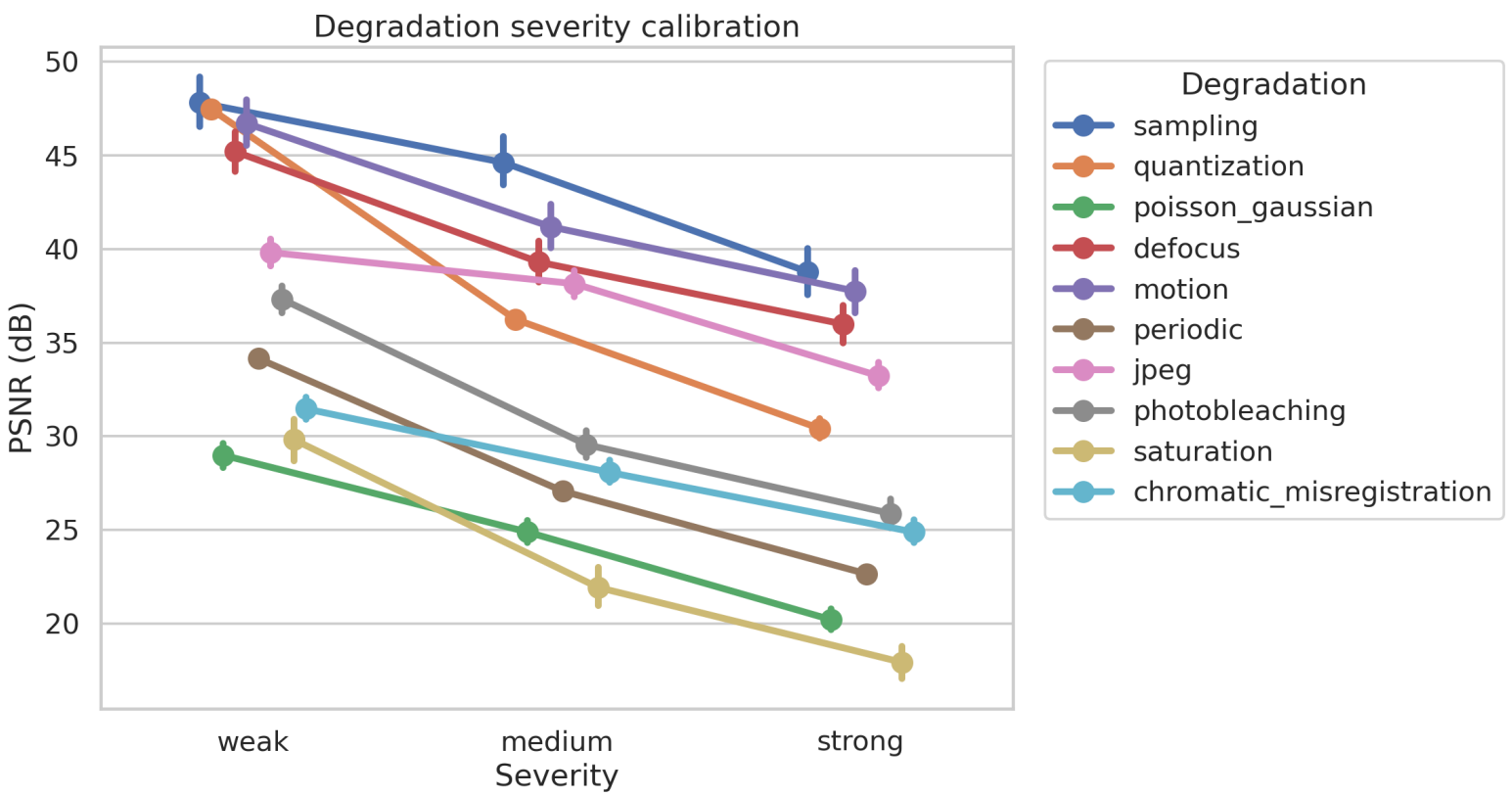
41.0%

severity accuracy · macro-F1 38.2%

Reading: classical features carry transferable signal, but modality/domain variation prevents a simple linear fingerprint from becoming a universal detector.

WEAK → MEDIUM → STRONG

Severity is empirically ordered across all ten families



99.95%

3,743 of 3,745 eligible paired triplets satisfy PSNR(weak) ≥ PSNR(medium) ≥ PSNR(strong).

Not equally spaced

Family-specific slopes show why severity must be calibrated per corruption rather than treated as one universal physical scale.

PAIRED INFERENCE-TIME ROBUSTNESS

One question; three pixel conditions; one causal comparison

Unique sample
image + official MCQ + one recipe

→

Clean
official Micro-Bench image

Degraded
exact Phase 1 recipe

Restored
matched restored image

Held constant

Question, answer options, expected answer, prompt template, model, decoding parameters, and sample-specific seed.

Changed deliberately

Only the image condition. Prompts never identify clean/degraded/restored; request order is deterministically shuffled.

Unit of analysis: the independent image–question triplet; no pseudoreplication from multiple questions per source image.

DETERMINISTIC, RESUMABLE EVALUATION

Balanced cohort → blinded requests → conservative scoring

01 · Join

recipes, generated, restored, and official annotations by recipe ID

02 · Balance

task, family, severity, modality; one question + recipe per image

03 · Shuffle

all three conditions interleaved to reduce order/load effects

04 · Infer

Qwen3-VL-8B via vLLM; deterministic JSON answer request

05 · Parse

JSON/letter/index/exact option only; ambiguity is never guessed

500	1,500	6	100%
UNIQUE PAIRED SAMPLES	COMPLETED REQUESTS	PERCEPTION TASK TYPES	PARSE RATE

Cohort: 167 coarse + 333 fine perception; 100 electron + 100 fluorescence + 300 light microscopy samples.

Report effects, confidence intervals, and transitions—not isolated accuracy

PRIMARY ENDPOINT

Task-macro accuracy

$$\mathrm{MacroAcc}(c) = \mathrm{mean}_{\mathrm{task}}\ \mathrm{mean}_{\mathrm{sample}}\ \mathbb{1}[\hat{y}_{i,c} = y_i]$$

Each task type receives equal weight, preventing frequent tasks from dominating.

10,000× bootstrap

Resample complete triplets; report paired 95% percentile intervals.

Exact McNemar

Test discordant pairs; Holm-adjust the three condition comparisons.

Four effects

Degradation loss, restoration gain, residual vs clean, recovery fraction.

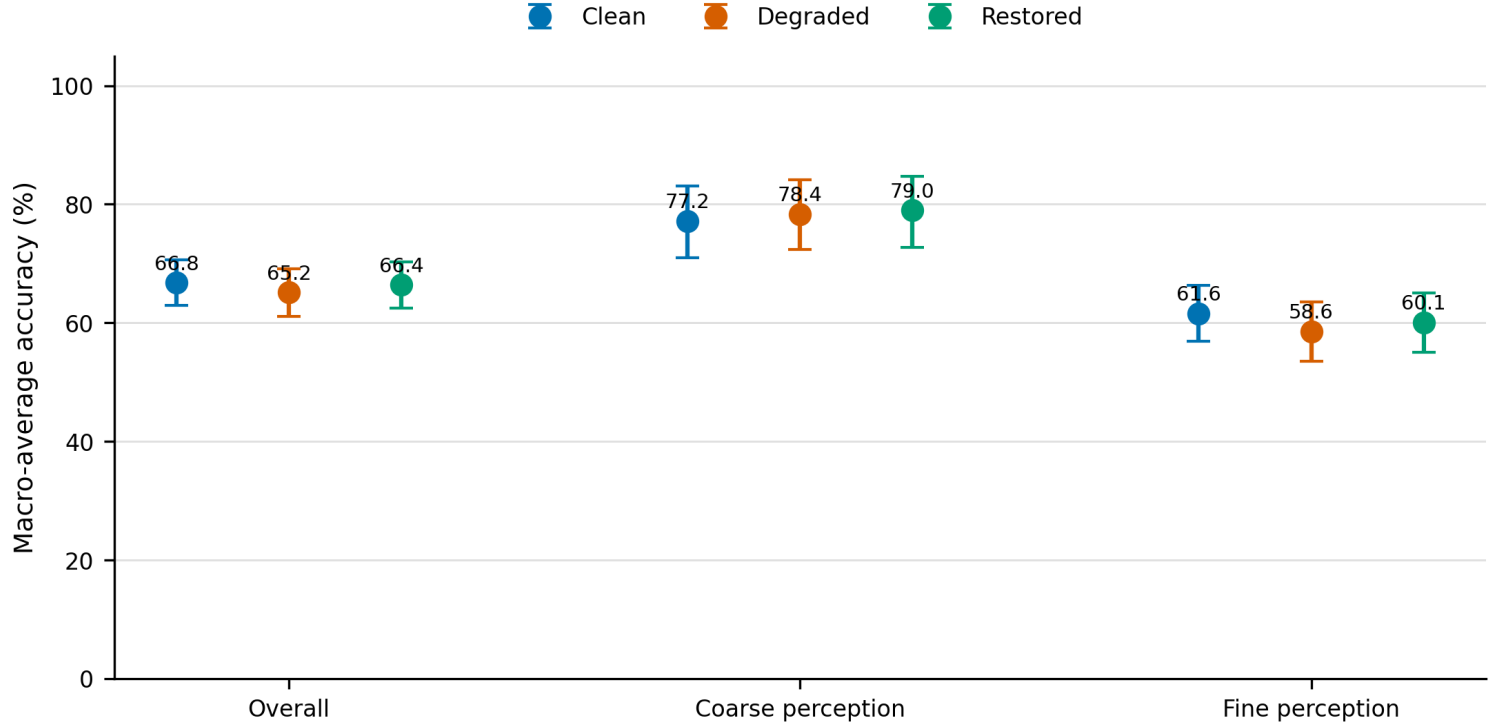
Transition audit

Stable, restored, unrecovered, harmed, helped, transient, only-restored, always-wrong.

Scope boundary: completed Phase 1 annotations support coarse and fine perception only; no cognition category is inferred or fabricated.

OVERALL ROBUSTNESS

Restoration recovers most—but not all—of the aggregate loss



Points are task-macro averages; bars are paired bootstrap 95% confidence intervals.

66.8 → 65.2 → 66.4%

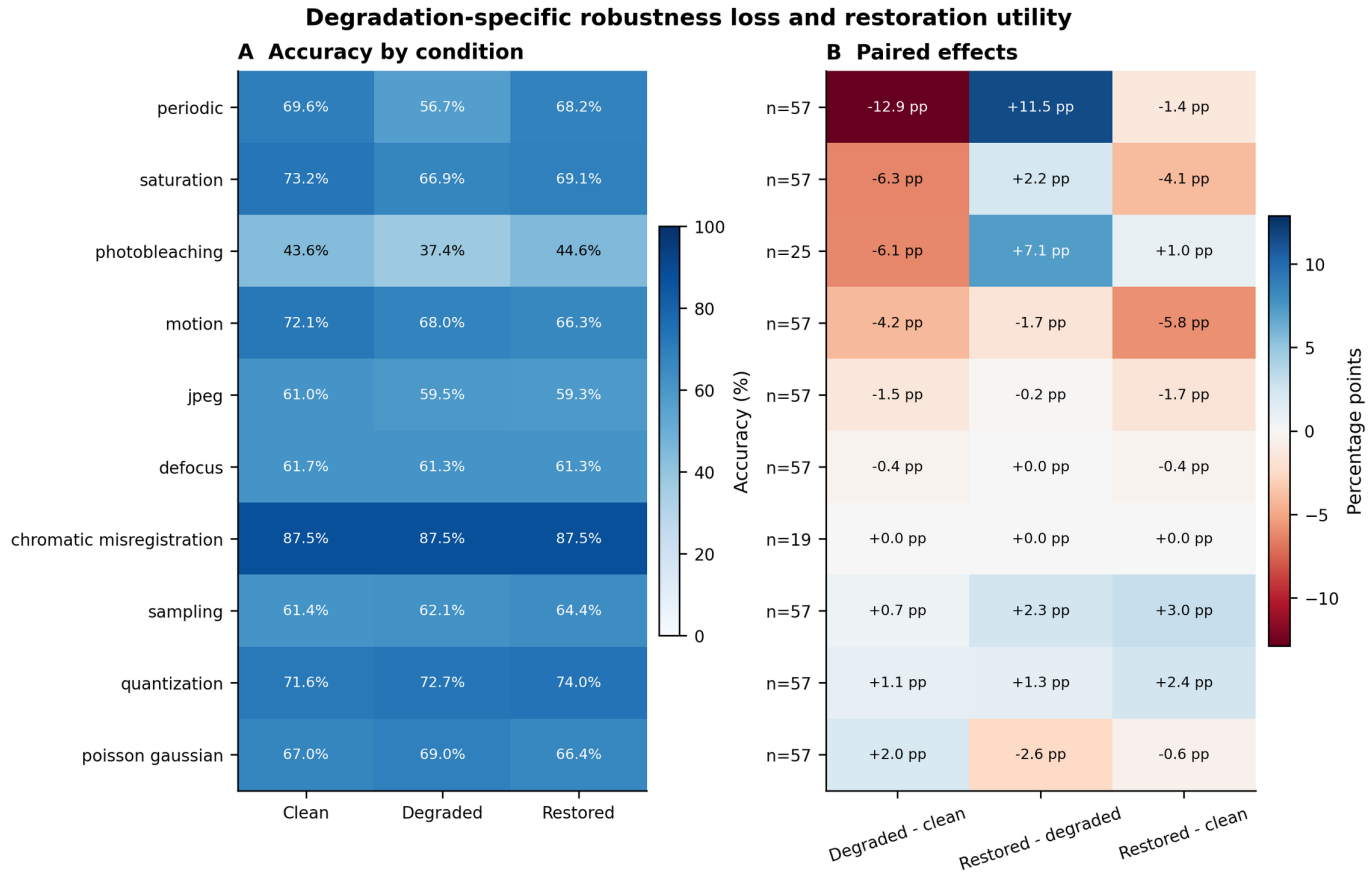
Overall macro accuracy: −1.62 pp degradation effect, then +1.21 pp restoration gain.

Fine perception

drops 3.02 pp and recovers 1.51 pp; coarse perception does not show the same loss. All 95% CIs overlap.

DEGRADATION-SPECIFIC RESPONSE

Aggregate averages hide the family that matters



Periodic: −12.9 / +11.5 pp

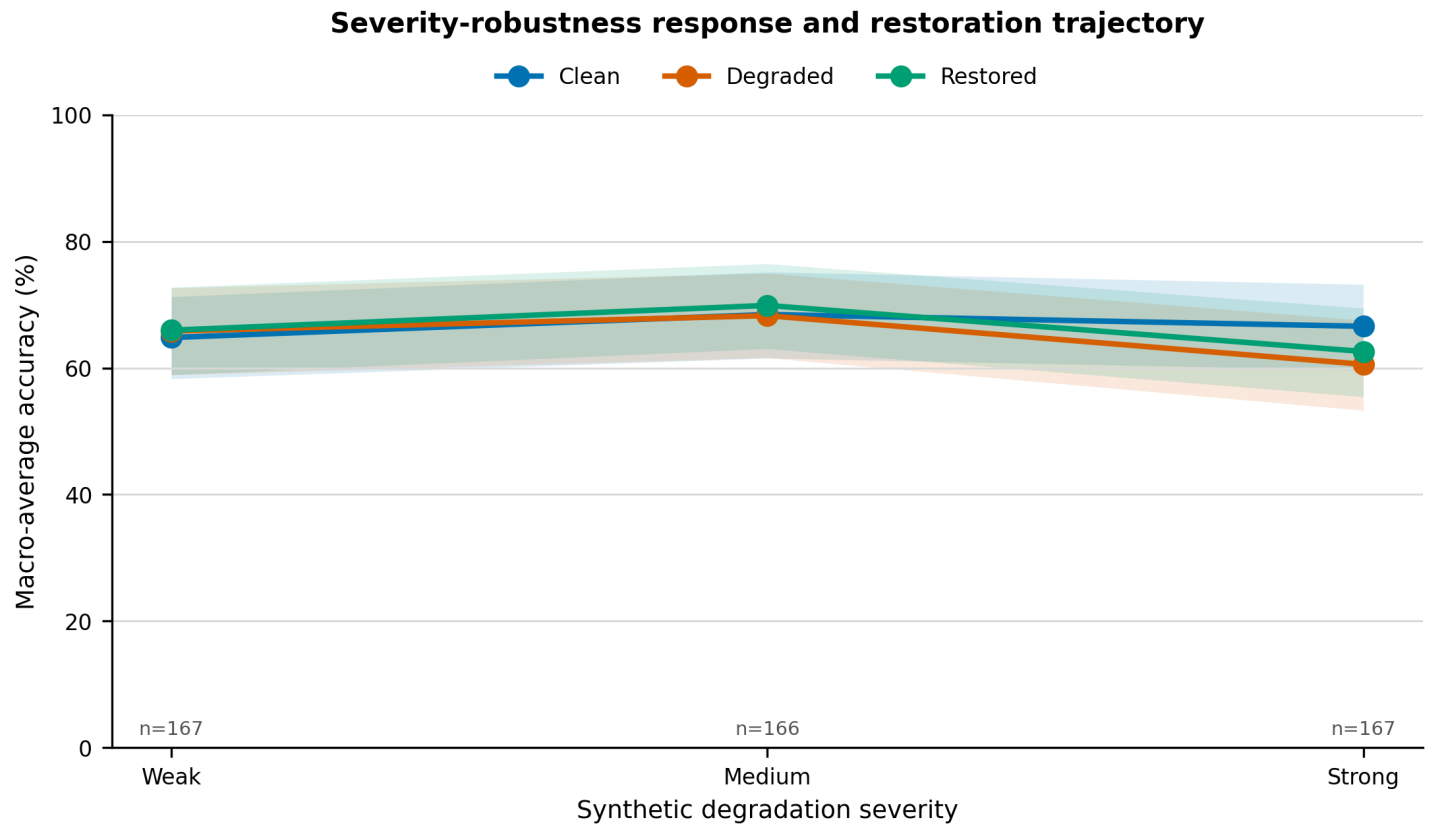
The largest robustness loss is nearly recovered by the known-frequency notch filter.

Heterogeneous utility

Photobleaching gains +7.1 pp, while motion restoration loses another 1.7 pp; subgroup n ranges 19–57.

SEVERITY RESPONSE

Strong corruption is where semantic robustness separates



Shaded regions are paired bootstrap 95% confidence intervals.

Strong: −6.0 pp

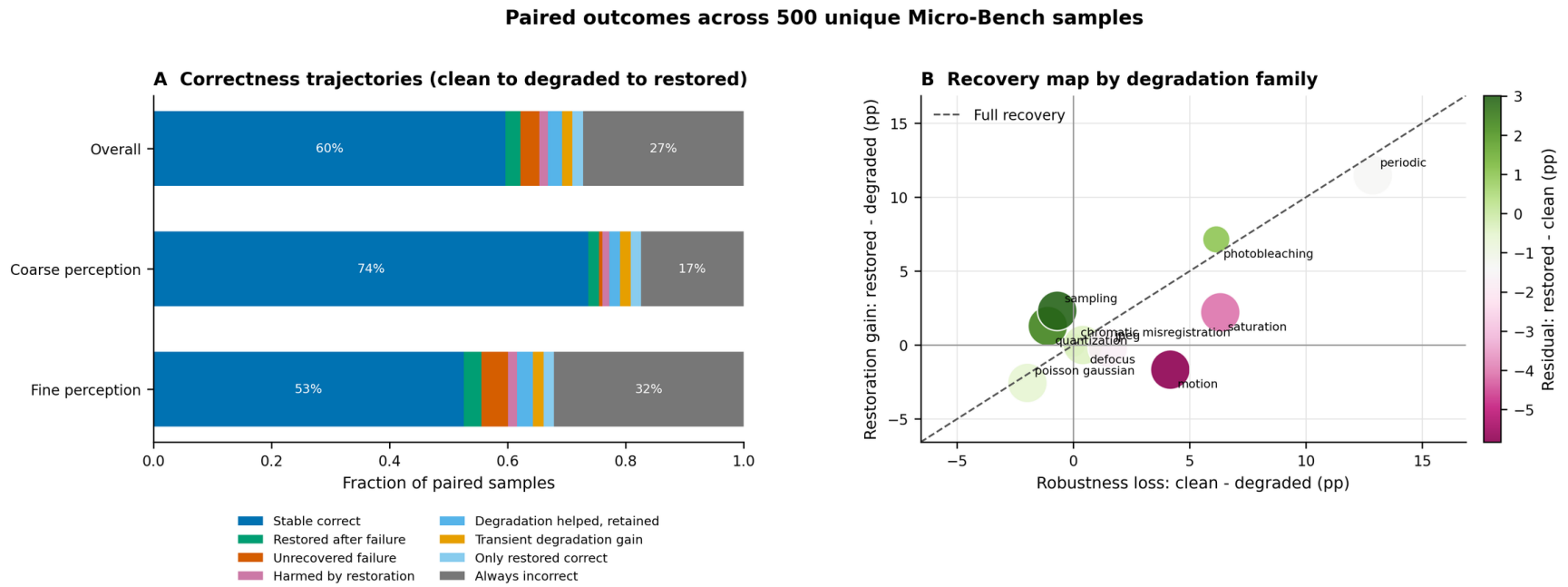
Clean 66.6% versus degraded 60.6%; restoration returns +2.0 pp, leaving −4.0 pp residual.

Weak ≈ unchanged

Weak and medium strata show no consistent loss. Curves use different balanced subsets, so all three baselines are shown.

PAIRED CORRECTNESS TRANSITIONS

Recovery must be traced sample by sample



59.6% stable correct

298 / 500 remain correct throughout; 136 / 500 (27.2%) are always incorrect.

13 restored · 16 missed · 7 harmed

Restoration reverses some degradation failures, but periodic—not motion—lies close to full recovery.

WHAT THE EVIDENCE SUPPORTS

Three conclusions—and one disciplined boundary

01
A reproducible DIP benchmark

1,701 clean sources become 15,480 controlled recipes with strict train/evaluation separation, full provenance, and 99.95% severity monotonicity.

02
Restoration follows invertibility

Notch filtering and denoising can recover structured/noisy corruption; clipping, quantization, sampling, bleaching, and compression remain information-limited.

03
Fine perception is the stress point

Overall loss is modest and uncertain, but strong corruptions and particular families expose meaningful paired failures that aggregate accuracy obscures.

Boundary: these are controlled synthetic degradations and exploratory n=500 VLM results; pixel improvement does not automatically imply biological or semantic improvement.

SHARIF UNIVERSITY OF TECHNOLOGY · COMPUTER ENGINEERING

Thank you.

Which image failures should a microscopy model be trained to notice—and when should it admit that information is gone?

Shayan Salehi · 404203509 | **Ali Hezaveh** · 404208531
Digital Image Processing Course · Prof. Kasaei